GROUP 1

MEMBERS :

NAME	Domain
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MAKOSSO Mignon	MIA

EEE DOMAIN

1. **Objectives of the domain**
2. **Notions of DC motors**
3. **Advantages and disadvantages of DC motor**
4. **Notions on the electronic components in the circuit**
5. **Block functional diagram of the circuit**
6. **Presenting the simulation with the code**

Animator: Abo William Junior

EEE DOMAIN

OBJECTIVES:

The electrical and electronic engineering (EEE) section of a boat project typically covers the design, installation, and integration of all the electrical systems and electronic components on the vessel.

During the first semester, as part of the electrotechnical boat project, electronic and automatic (EEE), Break down the motor control system into functional blocks, design and simulate the Analogic functional blocks model:

-Controlling the **speed** of the motor and the **direction** of rotation

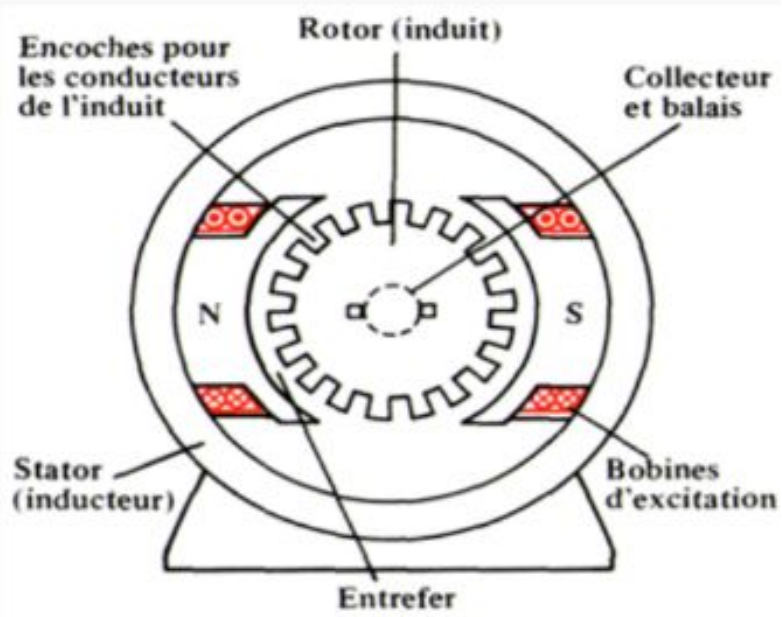
Let's first discuss on the Dc motor,

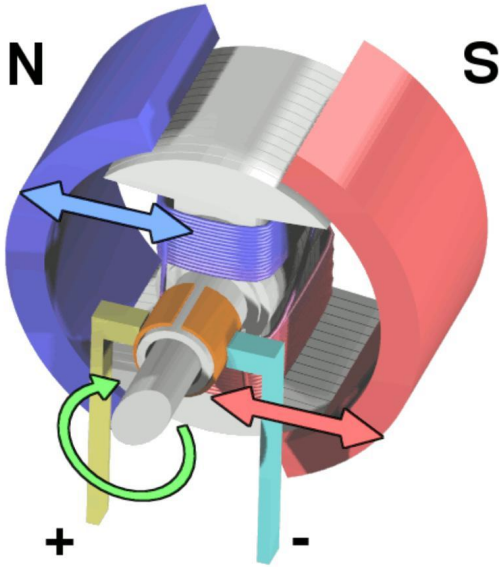
A DC motor is an electrical motor that uses direct current (DC) to produce mechanical force.The most common types rely on magnetic forces produced by currents in the coils. Nearly all types of DC motors have some internal mechanism, either electromechanical or electronic, to periodically change the direction of current in part of the motor.

ATTRIBUT	VALUE
Supply voltage	12 Vcc
Power range	36.88 W
Output speed	4289 tr/min
Shaft diameter	6.35 mm
Maximum output torque	82.08 gcm
Length	69mm
Dimensions	(∅)51.8*69 mm
Current	5.28 A
Core material	Steel Core

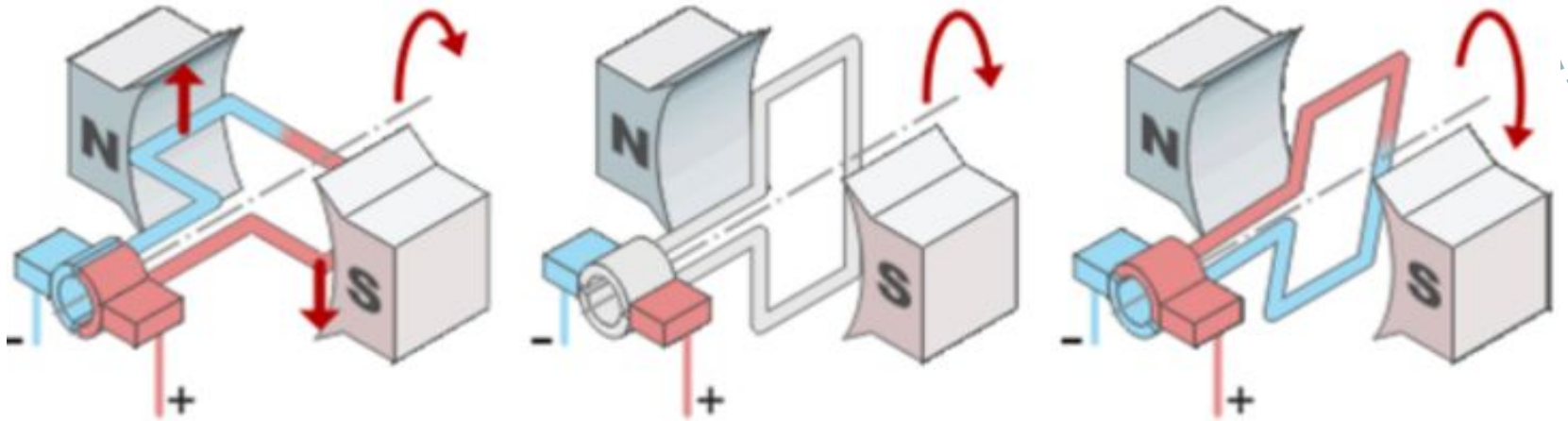
Principle of Dc motor:

A DC motor is made up of a magnetic circuit featuring a fixed part named stator, a rotative part named rotor and air-gap between both parts:The inductor (stator) creates fields which rotate by winding.An induced electrical circuit (rotor) is affected by this magnetic field. The collector and brushes provide access to the rotor circuit.

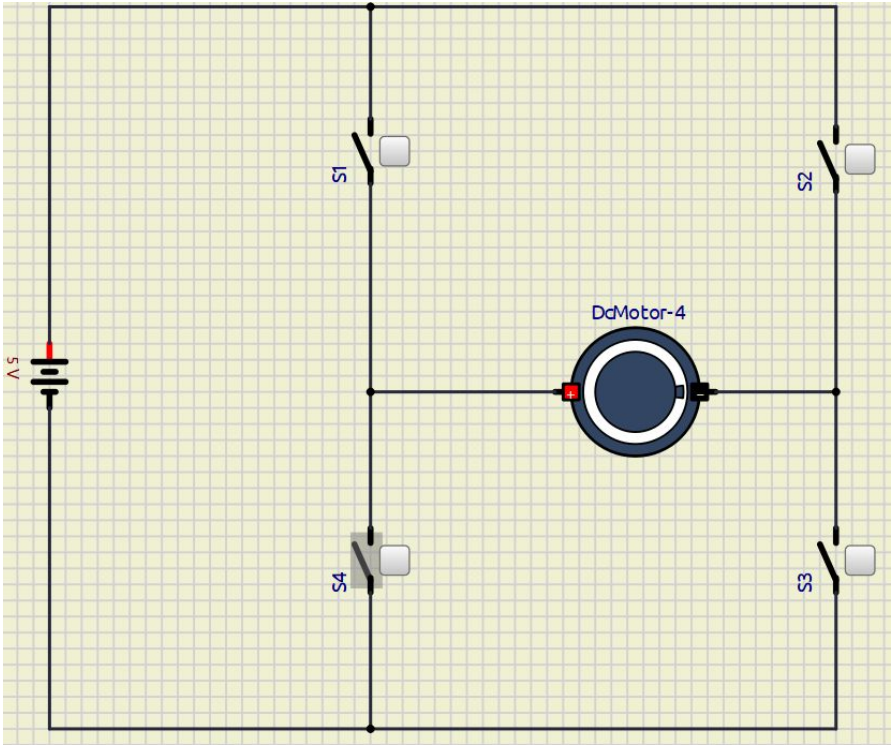


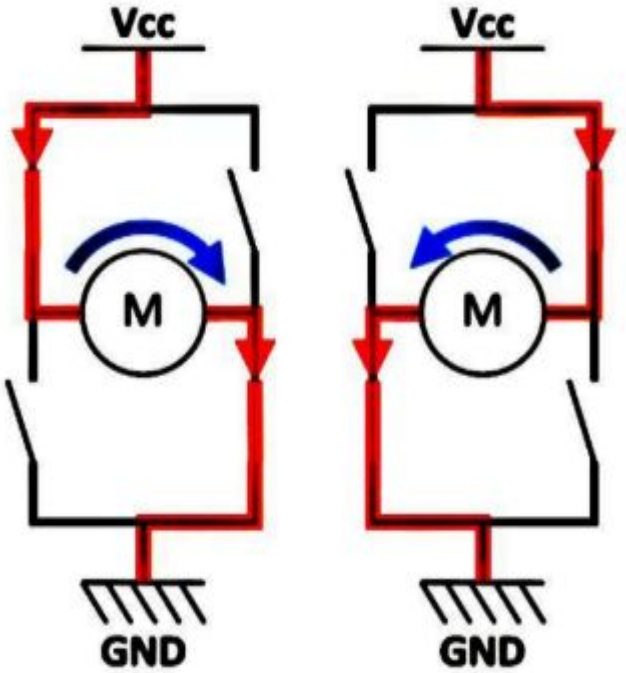


When the winding of a motor's inductor is powered by direct current, following the same principle as a permanent magnet motor (as shown in the figure below), it creates a North-South directional magnetic field (excitation flux). A coil capable of rotating about an axis of rotation is placed in the magnetic field. Additionally, the two conductors forming the coil are each electrically connected to a half-commutator and supplied with direct current via two brush contacts. According to Laplace's law (any conductor carrying a current and placed in a magnetic field is subject to a force), the conductors of the armature placed on either side of the brush axis (neutral line) are subject to equal but opposite forces F, producing a motor torque: the armature starts to rotate.



Controlling a motor to turn in 2 directions using switches, We observe that when S1 and S3 are closed(connected) while S2 and S4 are opened the motor turns towards the right direction and vice versa. It just function as an H-Bridge





Advantages of dc motor

1. DC motors have simpler installation and maintenance.
2. DC motors are typically more efficient and make better use of their input energy.
3. They have high startup power
4. They have high starting torque; it’s as high as 500% compared to normal operating torque.
5. They have fast response times to starting, stopping, and acceleration.

Disadvantages of dc motors

1. High initial cost.
2. Its operation cost and maintenance cost is very high due to the presence of commutator and brush gear.

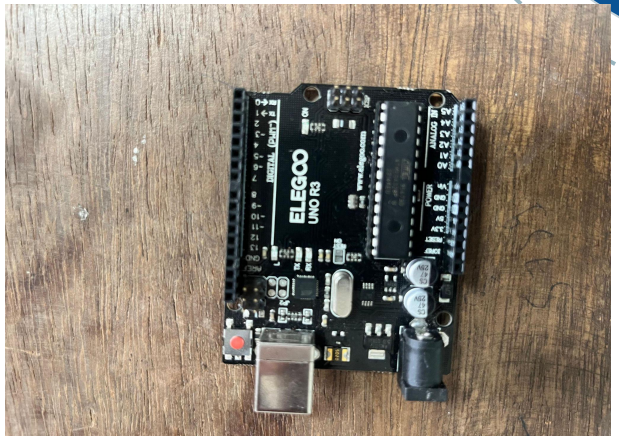


Other components present in the EEE Domaine

1-**Arduino microcontroller board**

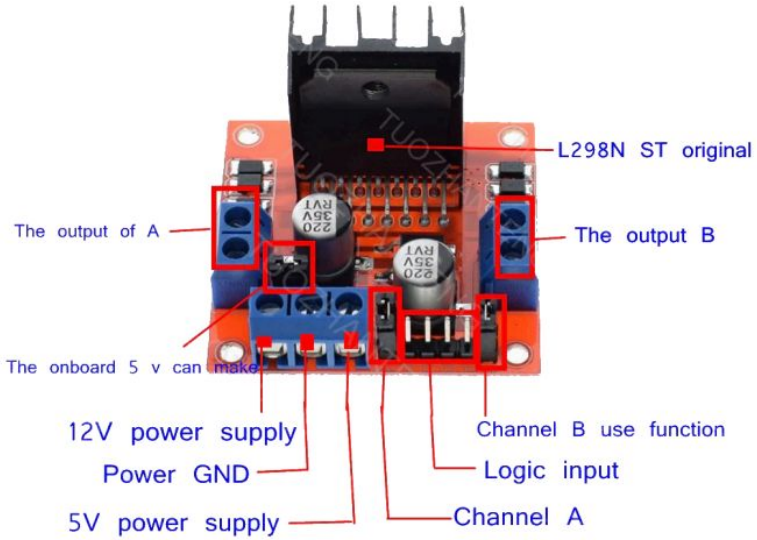
Arduino senses the environment by receiving input from many sensors, and affects its surroundings by controlling lights, motors, and other actuators.

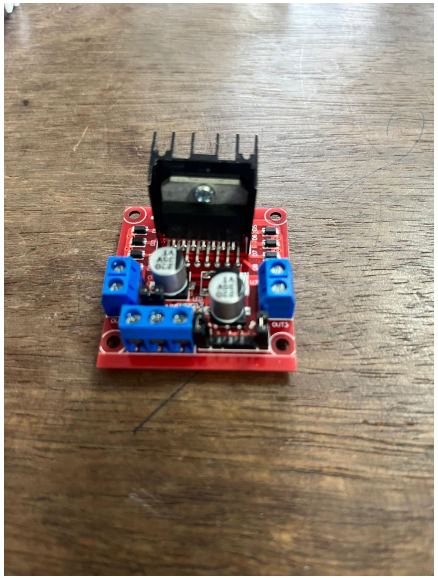




2-**DC Motor Driver(H-Bridge)**

The Dual DC Motor Driver 4A V3 allows two DC motors to be driven independently, controlling both speed and direction. The H-bridge receives the PWM signal as input and regulates the quantity energy supplied to the motor.





3-**Power**

It is essential to have an adequate source of energy to provide the voltage and current required for the operation of the H-bridge and the motor. This source can be a battery, a regulated power supply or any other source suitable energy.

4-**Button**

We've included a button to change the direction of rotation of our DC motor

5-**Potentiometer**

A user-adjustable 3 terminal variable resistor that can be used as a voltage divider. The first one will be used as a dimmer for the LCD screen, the second one will be used to set the boat direction.



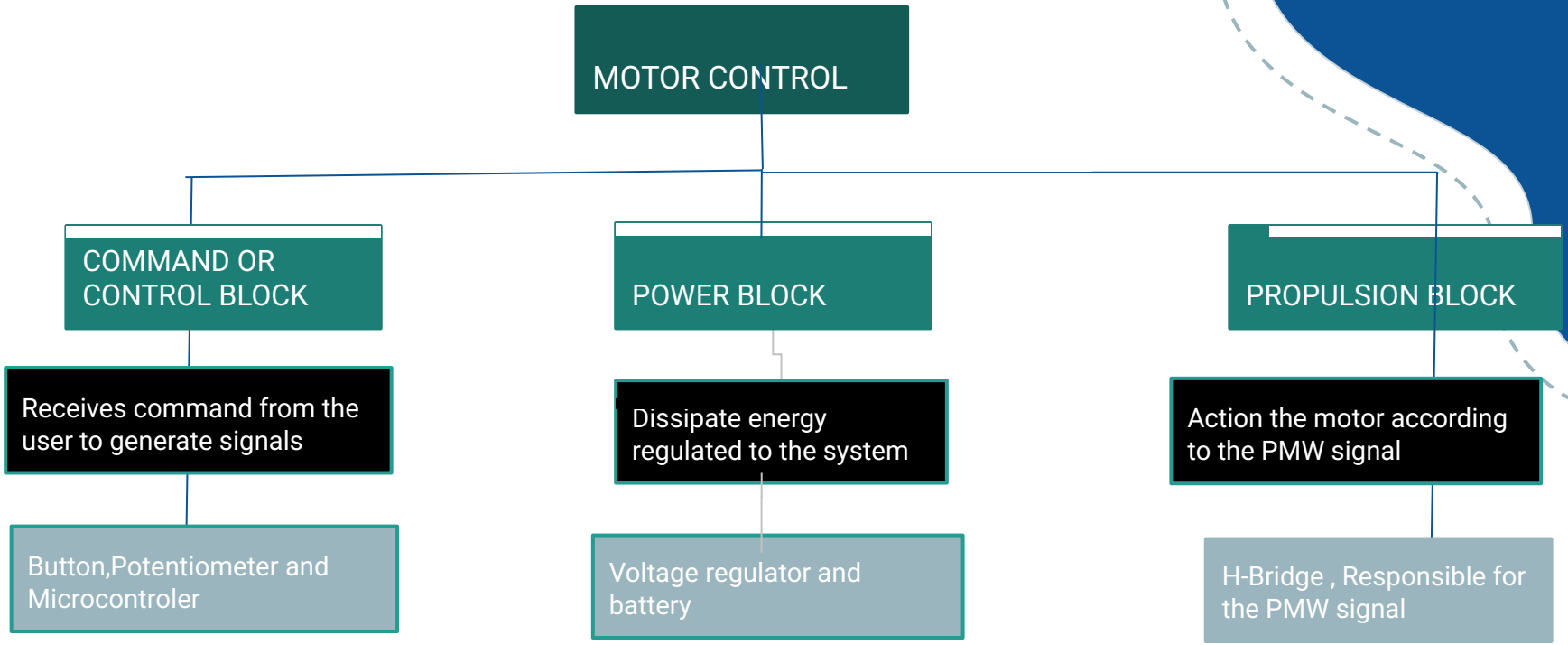


6-**Current and Speed Sensors**

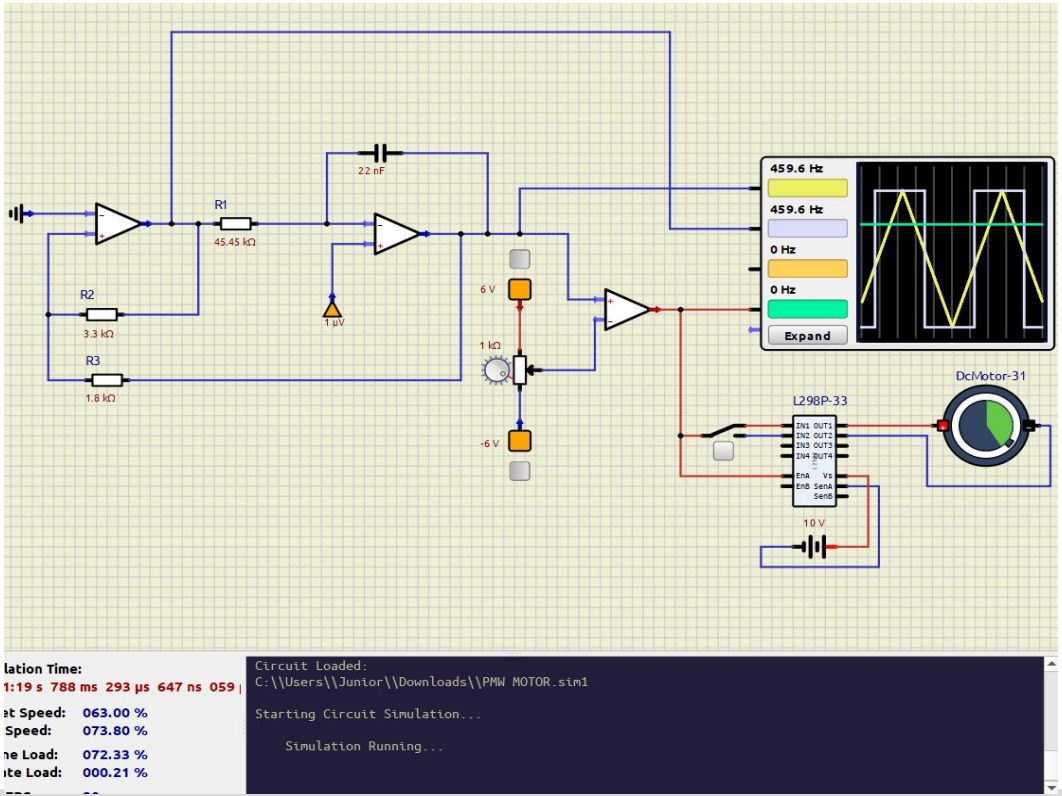
Speed sensor measures the rotational speed of the motor by detecting the movement of the motor shaft and generating a corresponding output signal, allowing the speed to be calculated.

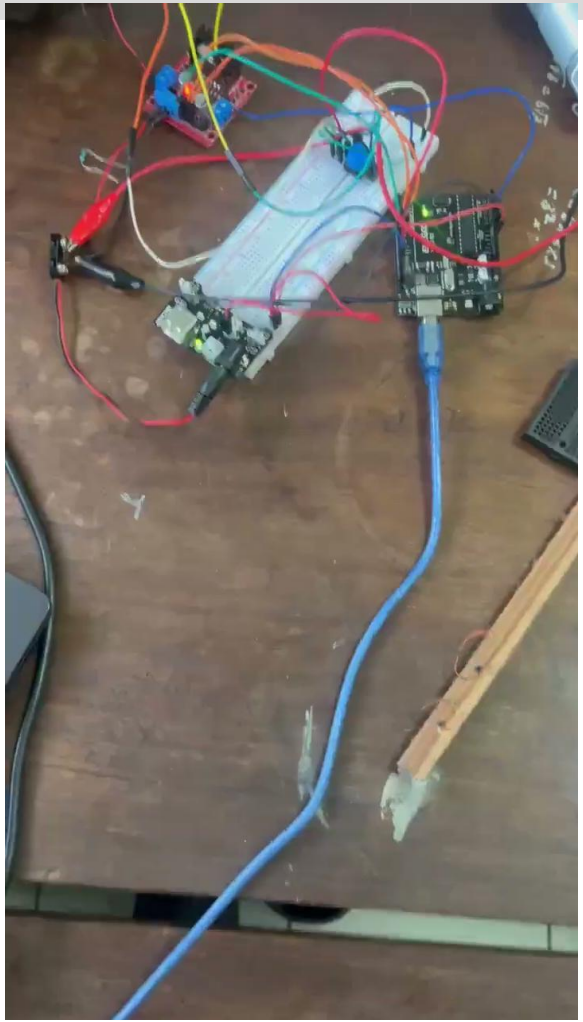
Current sensor using a sensor effect Hall, this device measures the current consumed by the motor. This information is essential for monitoring power consumption and preventing overcurrent.

BLOCK FUNCTIONAL DIAGRAM OF THE CIRCUIT



UNDERSTANDING THE SIMULATION OF PMW ON SIMULLIDE AND TEST IN PRACTICE



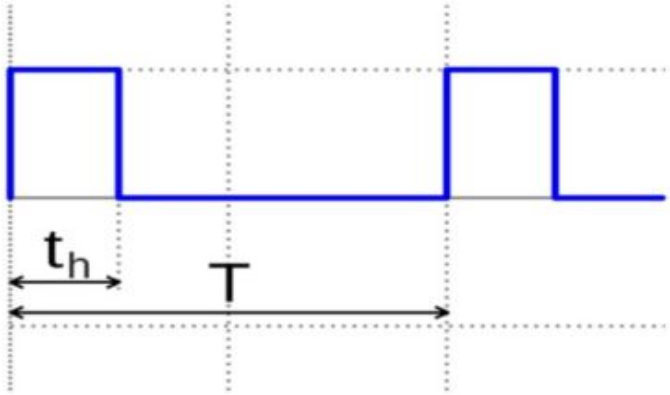


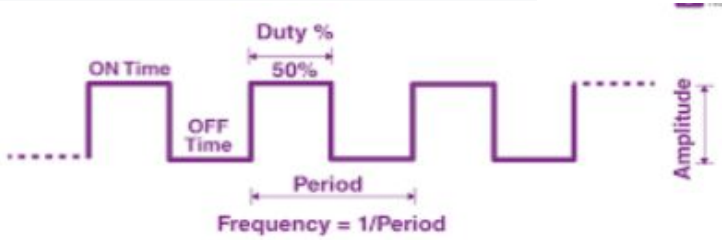
PMW SIGNAL(PULSE WIDTH MODULE)

PWM (Pulse Width Modulation) is a digital signal, therefore the voltage can only take two values. The signal is square. The low level generally corresponds to 0 Volts. The period is noted as T; the duration of the pulse (during which the voltage is that of the high state) is called th. More precisely, Pulse Width Modulation "PWM" provides a rectangular logic signal.

All the information resides in the duty cycle, which designates the duration of the high state during a period T. PWM varies the duration of the high state (which we will call th). This changes the duty cycle (the duty cycle is expressed as th/T and can vary from 0 to 1).

This rectangular signal, with a frequency of 12 Hz, has a duty cycle of 10%, which means that the electrical level remains in the high state for 10% of the total cycle time (one cycle - or period - lasting here for 1/12 of a second). Its average value is low compared to its maximum amplitude when it is in the high state.





Summary diagram and simulation